

Hybrid Information Retrieval System for Recipe Search Optimization

1. Purpose

The purpose of this project is to design and implement a hybrid information retrieval system (IRS) that improves the ranking and relevance quality of search results in the culinary domain. Recipe searching is a common information retrieval task where users often search for short and sometimes ambiguous queries for grocery shopping or meal prep ideas.

This hybrid IR system addresses the limitations of singular method based IR systems by combining lexical (BM25F) and semantic (Dense) retrieval techniques to deliver more accurate, context aware search results. Reciprocal Rank Fusion is also incorporated to combine the rankings of both algorithms into a unified set of results. Lastly, not as an enhancement to current algorithms but as an aesthetic choice, this system also exercises multimodal retrieval by also retrieving image data that corresponds to the each recipe document.

The goal is to enhance retrieval precision and improve overall user experience by having a system that accurately interprets a user query without demanding manual user query redefinition.

2. Background

Recipe retrieval systems are an essential part of digital food and wellness platforms.

Websites and apps like Yummly, Allrecipes, and Food Network have large collections of

recipes that users can submit and search queries to locate the relevant food content of their choice. These systems typically allow users to search by different fields such as dish category, ingredients, cooking time, dietary restrictions, etc. The current recipe search systems generally fall into two categories:

Traditional Keyword Based Systems

Websites such as Allrecipes rely mainly on keyword matching and manual user filtering. While this approach allows the IRS to run quickly, it notably struggles when user intent does not match exactly with the indexed terms.

For example, a user searching for “healthy quick meal” may expect recipes that are easy to prepare and nutritionally balanced. However, a strict keyword based system will determine which terms in the query have the most weight and prioritize documents containing exact matches for only those terms. For example, if the IRS chooses to weigh “quick” as the heaviest term of importance then the results will be meals that are quick to prepare, but they may also be high in calories, thus negating the intent in the rest of the query.

Semantic Based Systems

Other platforms such as Yummly focus on semantic searching for their IRS. While this approach captures meaning more effectively for conceptual user queries, it still may miss exact keyword matches and retrieve more generally related results. For example, a user may do an ingredient specific search such as “banana bread with walnuts” but a purely semantic system may return more broadly similar results such as banana muffins with

cashews, walnut cakes, or general fruit and nut desserts, rather than enforcing all the key word constraints together.

3. Problem Statement

The core problem this hybrid system addresses is the inability of recipe retrieval systems that focus solely on one retrieval method to effectively interpret user queries and provide relevantly ranked results. Some user frustrations that manifest from this issue are poor relevance, ranking errors, or vocabulary mismatches.

Poor relevance stems from the disconnect between semantic understanding and lexical interpretation of keywords. As a result, users become frustrated, needing to repeatedly refine or rephrase their queries to obtain the intended results instead of finding them in a single search. Ranking errors occur when relevant documents are retrieved but fail to appear in top positions. Because users typically engage with only the first few results, poor ranking significantly reduces usability. Vocabulary mismatches are when users and documents use different terms to express the same concept, for example, “quick” vs “fast”.

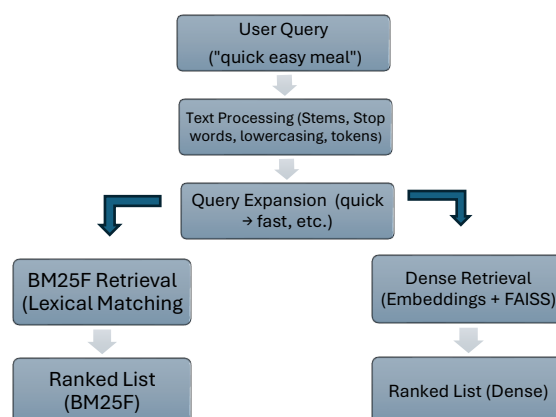
In a business setting, these issues can reduce user engagement, session duration, and user return visits.

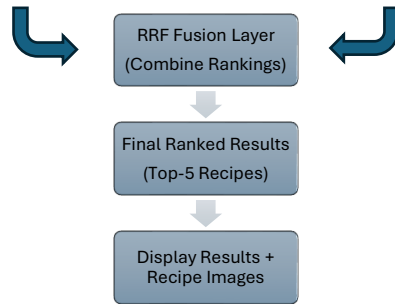
4. Solution

To address these challenges, a hybrid IR System was developed that integrates enhanced lexical (field weighting) and semantic retrieval, query expansion, and rank fusion. This system was built with the use of corpus of over 4000 recipes, where each recipe

(document) has multiple fields: title, ingredients, instructions, and Image metadata. The documents (and subsequent queries) all undergo text processing techniques such as stop word removal, stemming, lowercasing, and tokenization. The system then incorporates a small rule based query expansion to bridge the vocabulary gaps in some common recipe conceptual terms such as “quick” and “fast” or “easy” and “simple.”

Next the system uses BM25F as the primary keyword matching retrieval method while also assigning weights to the different recipe fields. The title and the ingredients were assigned higher weights, and the instructions were assigned lower weights because the instructions consisted of a lot of noise and common, low importance text. In parallel, the system also uses Dense Retrieval with the use of a transformer based encoder to capture semantic meanings in the documents. The queries are also encoded and matched based on cosine similarity scores with the use of the FAISS index. Lastly, the ranked results from BM25F and Dense Retrieval are combined through Reciprocal Rank Fusion to simply balance and merge the different scoring scales of each algorithm.





Now the system will return the top 5 ranked results per query along with their scores, title, ingredients, instructions, and image. So, in addition to the text based retrieval, the system also incorporated a lightweight multimodal component by linking each recipe document to its corresponding image based. This is done through a matching process that aligns the retrieved document's image metadata field with its image metadata stored in a folder apart from the corpus.

5. Results & Benefits

The system was evaluated using manually labeled relevance judgments across 10 queries. The hybrid system performed strongly on both performance measures that were calculated: P@5 and nDCG@5. It achieved a Precision@5 score of 0.86, indicating that on average about four out of the top five retrieved documents are relevant. Also, the system achieved a nDCG@5 score of 0.9231, which demonstrates strong ranking quality that captures semantic relevance beyond simple matching. Due to the nature of nDCG this high score indicates that the system is not only retrieving relevant results but is also consistently ranking the most relevant documents near the top positions. Overall, this hybrid system performs very well for this domain and will improve user experience and retention by allowing faster decision making from accurate results.

6. Risks and Mitigations

One risk with deploying the IRS is that the dense retrieval component adds significantly more computational cost to the system than a traditional lexical method system. To mitigate this, FAISS is incorporated to create more efficient indexing to minimize slowness. Another potential risk is latency from the query expansion component by introducing noise to the system. To mitigate this, I introduced only a short, controlled list of expansion rules that are the most pertinent to the culinary domain.

7. Conclusion

This hybrid information retrieval system has demonstrated a successful implementation of combining algorithms and retrieval techniques to improve the performance of IR systems in the culinary domain. By combining BM25F, dense retrieval, query expansion, and reciprocal rank fusion, this system has achieved strong relevance and ranking performance in both P@5 with a score of 0.86 and nDCG@5 with a score of 0.9231. The system also successfully addresses key limitations of traditional and single method searches while also navigating potential latency disruptions from the added complexity of the dense retrieval and query expansion components. For aesthetic and user enjoyment, the system also performs a simple multimodal image retrieval that matches the corresponding recipe images to each retrieved document. In the long term this hybrid retrieval system will improve user satisfaction and retention.